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lovefrequency528 | Arduino 1.8.19
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lovefrequency528

int numb=947; // if no switch is pressed, play the 528Hz tone
void setup() {

  pinMode(13, OUTPUT);
  pinMode(2, INPUT_PULLUP);
  pinMode(3, INPUT_PULLUP);
  pinMode(4, INPUT_PULLUP);
  pinMode(5, INPUT_PULLUP);
  pinMode(6, INPUT_PULLUP);
  Serial.begin(9600);
  pinMode(7, INPUT_PULLUP);
  pinMode(8, INPUT_PULLUP);

  int freq1 = digitalRead(2);
  int freq2 = digitalRead(3);
  int freq3 = digitalRead(4);
  int freq4 = digitalRead(5);
  int freq5 = digitalRead(6);
  int freq6 = digitalRead(7);
  int freq7 = digitalRead(8);

  if (freq1==LOW)
    numb=1262;
  if (freq2==LOW)
    numb=1199;
  if (freq3==LOW)
    numb=974;
  if (freq4==LOW)
    numb=782;

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Fig. 3: Screenshot of source code

square wave has DC component of half the value, which is blocked by the series capacitors C3 and C7 at the output. Audio mini module LM386 or PAM8403 has an internal capacitor and so no external part is needed when using it.

The square wave is applied to an amplifier and loudspeaker. Since the speaker must be of good quality, the author used the WS-887 mini speaker unit, which costs less and gives good sound quality.

The selection of the frequency is done by grounding the digital input pins using switches S1 through S6 shown in the circuit diagram (Fig. 2). These switches are normally open and are selectively grounded to choose the frequency.

The frequencies are generated as square waves, which are output from digital pin number D13. This square wave can be filtered to remove the harmonics, if necessary; but it is more wholesome with the harmonics. After all, the 528Hz square wave will comprise the third and fifth harmonics at 1584 and 2640Hz, whose amplitudes will be one-third and one-fifth of the main 528Hz. So, the square wave can be used even without a filter. The same

PARTS LIST

<i>Semiconductors:</i>	
Board1	- Arduino uno
IC1	- LM386 low-power amplifier
<i>Resistors (all 1/4-watt, ±5% carbon):</i>	
R1	- 1.5-kilo-ohm
VR1	- 4.7-kilo-ohm pot
<i>Capacitors:</i>	
C1	- 47µF, 16V electrolytic
C2	- 220µF, 16V electrolytic
C3, C5	- 100µF, 16V electrolytic
C4	- 10µF, 16V electrolytic
C6, C7	- 100nF ceramic disk
<i>Miscellaneous:</i>	
CON1	- 2-pin connector
S1-S7	- On/off switch
LS1	- 4-ohm, 0.5-watt speaker

applies to all the other frequencies.

The Arduino program just uses the microsecond delay function to generate the square wave. For example, the 528Hz frequency has a period of 1894 microseconds, because $1/528 \times 1000000 = 1893.9$. This is the time period for one cycle. One half cycle of the wave is positive 5V and the other half cycle is 0 volt. For half this time pin 28 (D13) is made high and for the other half low.

For the main 528Hz frequency no digital pin (D2 through D7) is grounded via switches. For the other frequencies, digital pins D2 through D7 are grounded one at a time and power is applied. The corresponding frequency tone is output from the same pin number 28 (D13).

When no switch is grounded, only the love frequency 528Hz is output. It is necessary to select the switch before power is given to the board, as it is not possible to change the tone when a tone is already being played. That is also inadvisable because tones should not be changed often; one must hear a particular tone for a considerable time for it to be effective.

Software

Arduino IDE is used to write the sketch (software). The program is uploaded onto the Arduino board from the computer running the Arduino IDE. Thereafter, the Arduino can be operated from a separate 5V supply or a 9V flat battery pack connected to the 9V socket on the board. A screenshot of the source code is shown in Fig. 3.

Construction and testing

First of all upload the source code lovefrequency528.ino into the Arduino Uno after selecting proper board and port from Arduino IDE. Disconnect the board from computer and connect 5V across CON1. Now assemble the circuit on a breadboard.

You may use a readymade LM386 amplifier, as shown in dotted box, or wire it up as per the circuit diagram. Connect points B and C to the amplifier board if the board is using an external power supply. If the board is not using any external power supply, connect points A, B, and C from Arduino Uno to the amplifier.

After assembling the circuit on PCB, enclose it in a suitable box. The author used WS-887 mini speaker in his project. In EFY lab, the LM386 low-power amplifier was assembled on a general-purpose mini zero PCB, as shown in Fig. 1.

To test the usefulness of the tones, you should hear them for about 30 minutes at least.

EFY note. If you have a frequency meter, you may check the frequency and adjust it to get 528Hz, as near as possible. In EFY lab, we checked only the generation of frequency and not its impact on human beings. **EFY**

EFY Note

The source code of this project is available

for free download at www.electronicshobby.com



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